MTH 231(C) Dr. James Ashe
Exam 2 Review/Quiz Due November 11, 2012 **Name:**

6.1: Velocity and Net Change

HW problems 7-11 (odd). *Find the displacement, and find the distance traveled over the given interval.*

1. $v(t) = 10 \sin 2t$; $0 \leq t \leq 2\pi$

HW problems 12 and 15. *Determine the position function (you may use EITHER the FTC or anti-derivative method), and determine when the object is moving in the positive and negative directions.*

2. $v(t) = \dfrac{20}{\sqrt{t+1}}$ on $[0, 8]$; $s(0) = -4$ (use substitution)

HW problems 18, 25, 26, 31, and 32. *Word problems applying the same concepts as above.*

3. A culture of bacteria in a petri dish has an initial population of 1500 cells andgrows at a rate (in cells/day) of $N'(t) = 200(t+2)^{-1/2}$.
  a. What is the population after 14 days?
  b. Find the population $N(t)$ at any time $t \geq 0$.

6.2 Region Between Curves

HW problems 5-6, 10-11, 15-18, 23-24 *Find the area of the region.*

4. The region bounded by $y = x$ and $y = x^2 - 2$.

5. The region in the first quadrant bounded by $y = (x-1)^3$ and $y = x - 1$.

6. The region bounded by $y = x$ and $x = (y-2)^2$.

6.3 Volume by Slicing

HW problems 7 and 9. *Use the general slicing method to find the volume of the following solids.*

7. A triangular wedge whose perpendicular sides have lengths 3, 4, and 5 (use calculus; see the figure on page 343).

8. The solid with a semicircular base of radius 5 whose cross sections are perpendicular to the base and parallel to the diameter.

HW problems 15-18. *Let R be the region bounded by the following curves. Use the disk method to find the volume of the solid generated when R is revolved around the $x-$axis.*

9. $y = 2x$, $y = 0$, $x = 3$ (see the figure on page 344).

10. $y = 4 - x^2$, $y = 0$, $x = 0$ (see the figure on page 344).

11. $y = \cos x$, $y = 0$, $x = 0$ (Recall that $\cos^2 x = \dfrac{1}{2}(1 + \cos 2x)$.)

6.5 Length of Curves

HW problems 3-9 odd. *Find the arc length of the following curves on the given interval by integrating with respect to x.*

12. $y = \dfrac{1}{3}x^{3/2}$; $[0, 60]$

13. $y = \dfrac{(x^2+2)^{3/2}}{3}$; $[0, 1]$

14. $y = \dfrac{x^4}{4} + \dfrac{1}{8x^2}$; $[1, 2]$